
Denoising Diffusion Probabilistic Models for Source Camera Identification in Image Forensics

Zahra Farzadpour¹, Farah Nafees Ahmed¹, Fouad Khelifi¹

¹School of Computer Science, Northumbria University

`zahra.farzadpour@northumbria.ac.uk`

`farah3.ahmed@northumbria.ac.uk`

`fouad.khelifi@northumbria.ac.uk`

Abstract: Source Camera Identification (SCI) is a forensic task to determine the physical imaging device for a given digital image or video. The problem has raised concern with the proliferation of digital cameras in consumer devices, the ease with image capturing, editing, and sharing at scale across social media and messaging platforms, and the operational need to attribute illicit content in important evidence cases like child abuse to specific recording devices in criminal investigations. The main signal for SCI is Photo-Response Non-Uniformity (PRNU), classically extracted via a denoising filter such as the Wiener filter, which degrades substantially on small image patches. We propose a combined framework in which a per-camera Denoising Diffusion Probabilistic Model (DDPM) acts as a camera-dependent residual extractor, replacing the classical filter, and Linear Discriminant Analysis (LDA) exploits the full NCC score vector across all candidate cameras for the final identification decision. Evaluated on three benchmarks at 128×128 patches, our method achieves macro-averaged balanced accuracies of 93.74%, 93.84%, and 92.50% on the Northumbria, Dresden, and VISION datasets respectively, outperforming the Wiener filter on all three benchmarks.

Keywords: Source Camera Identification, Image Forensics, Denoising Diffusion Probabilistic Model, Linear Discriminant Analysis.

1. Introduction

Source Camera Identification (SCI) is a central forensic task, determining which physical imaging device captured a given digital image [1, 2]. The problem has grown increasingly important with the proliferation of digital cameras in consumer devices and the ease with which images are captured, edited, and shared at scale across social media and messaging platforms. In forensic investigations, it is often necessary to identify which device captured a piece of harmful content, such as child sexual abuse imagery, harassment material, or evidence in criminal cases. [3].

The most common signal used for SCI is Photo-Response Non-Uniformity (PRNU) [4, 5], which arises from small imperfections introduced during manufacturing in the photosites of a Complementary Metal-Oxide-Semiconductor (CMOS) or Charge-Coupled Device (CCD) sensor. Each pixel exhibits a slightly different sensitivity to light, producing a multiplicative gain pattern that is fixed for a given sensor across all images it captures. The traditional PRNU-SCI pipeline applies a denoising filter F to obtain a residual $R(I) = I - F(I)$, averages many such residuals into a per-camera fingerprint, and matches queries to camera fingerprints via Normalised Cross-Correlation (NCC). The Wiener filter, wavelet-based MAP denoising [6, 7], and Block Matching 3 Dimensional (BM3D) [8] filtering are common options for F . The Lukáš et al classical baseline [4, 5] is formed by the Wiener filter. A substantial body of work has refined this pipeline [9–15],

yet the core architecture — a hand-crafted denoiser feeding a residual correlator — has remained essentially unchanged.

A parallel body of work has replaced the hand-crafted denoiser with a learned representation. NoisePrint [16] trains a Siamese CNN to produce a model-discriminative camera fingerprint by suppressing scene content; subsequent work combines NoisePrint with classical PRNU to improve device-level identification [17]. Other deep-learning approaches train classifiers directly on residual or pixel patches, including CNN-based SCI [18–21], constrained convolutional networks [22], and recent dual-branch and fusion-based architectures [23–26]. Other research in this lineage has investigated CNN, hierarchical-fusion, and vision-transformer formulations of SCI on both image and video data [27–30].

To date, however, no work has investigated *Denoising Diffusion Probabilistic Models* (DDPMs) [31–33] as the denoising stage of a PRNU-style SCI pipeline. This is a natural question: a DDPM is, by construction, a content-prior denoiser, and recent work in image forensics has shown that a diffusion model’s reconstruction residual carries information discriminative enough to separate real from generated imagery (DIRE [34]). If a diffusion model trained on a single camera’s images learns a content prior that excludes that camera’s specific noise pattern, then the partial-noise / partial-denoise residual should retain device-specific information — much as the classical Wiener residual does, but produced by a denoiser whose inductive bias is learned from data rather than hand-designed.

A particular challenge motivating this work is source camera identification from small image patches. In practical forensic scenarios, the image available to the investigator may be distorted, partially corrupted, or only a small region may remain usable, for instance, a cropped thumbnail, a social-media-compressed fragment, or a single video frame. Classical PRNU-based SCI relies on averaging device noise over large pixel areas to achieve a reliable signal-to-noise ratio; performance degrades substantially as patch size decreases [15, 17]. Operating at 128×128 patches, our framework is designed specifically for this challenging regime.

Contributions. In this work we develop and analyse a per-camera DDPM-SCI pipeline evaluated on three benchmarks (Northumbria, Dresden, and VISION) at 128×128 patches. Our contributions are:

- We propose the first per-camera DDPM framework for SCI at small patch sizes (128×128), where conventional PRNU-based methods become unreliable. The DDPM acts as a camera-dependent residual extractor whose learned content prior suppresses scene variation while retaining device-specific noise patterns, addressing the signal degradation that limits classical denoisers on small patches.
- We show that Linear Discriminant Analysis applied to the full NCC score vector consistently improves identification accuracy across all three datasets, by exploiting inter-camera relationships that single-score matching discards.

The remainder of the paper is organised as follows. Section 2 reviews PRNU-based and learned-residual SCI, together with diffusion models for image forensics. Section 3 describes the proposed pipeline, the PRNU-guided noise objective, and the residual extraction and matching procedure. Section 4 presents three benchmark datasets, result analysis. Finally, the paper concludes in section 5.

2. Related Work

Related work falls into three areas: classical PRNU-based SCI, learned-representation SCI, and diffusion models for image forensics.

2.1. Classical PRNU-based SCI

Lukáš, Fridrich, and Goljan formalised the conventional PRNU-SCI pipeline [4, 5], expanding on the multiplicative sensor-noise model where each photosite contributes a fixed gain factor that varies over the sensor surface. The pipeline estimates this gain pattern by averaging denoising residuals across many images, then matches query residuals to per-camera fingerprints via Normalised Cross-Correlation. Subsequent work has improved the denoiser, the fingerprint, or the matcher independently [9, 35].

The denoiser stage has received the most attention. Mihçak et al. [6] introduced spatially-adaptive wavelet denoising, which remains the standard reference filter. The fingerprint construction stage has been refined

via content removal [9] and the multiplicative-model treatment of Fernández-Menduiña et al. [14]. Tian et al. [15] specifically address PRNU extraction from highly-compressed low-resolution blocks, sharing the operating regime of our work. The classical pipeline remains highly effective on full-resolution images [35], but degrades substantially as patch size decreases [15].

2.2. Learned-representation SCI

A parallel line of research replaces the hand-crafted denoiser or the correlation-based matcher with a learned representation. Early CNN-based SCI [18,19] trained classifiers directly on small image patches, demonstrating that learned features can outperform classical hand-crafted statistics at patch level. Bayar and Stamm’s constrained convolutional network [22] introduced a first-layer high-pass-like constraint that suppresses image content while preserving forensic traces.

Cozzolino and Verdoliva [16] introduced NoisePrint, a CNN-based camera *model* fingerprint trained with a Siamese objective on same- and different-camera patch pairs, later combined with classical PRNU for robust device-level identification [17]. Our work shares NoisePrint’s conviction that the denoiser stage benefits from learned inductive biases, but uses a generative content-prior model (a DDPM) trained per camera rather than a single shared discriminative encoder.

Recent work has developed CNN, hierarchical-fusion, and vision-transformer formulations of SCI on both image and video data [23,27–30]. None of this work, to our knowledge, has investigated diffusion models as the learned denoising stage of a PRNU-style pipeline.

2.3. Diffusion models for image forensics

Denoising Diffusion Probabilistic Models [31,33,36] model a data distribution through an iterative denoising process trained to reverse a fixed Gaussian forward corruption. The DDIM sampler [32] accelerates inference by recasting the generative process as a deterministic ODE, reducing the number of function evaluations from ~ 1000 to ~ 50 at no measurable quality cost. Diffusion models have rapidly become the dominant generative paradigm for natural images, with applications ranging from unconditional synthesis to image restoration, inpainting, and super-resolution.

Within image forensics, diffusion models have been used primarily as a detection target rather than as an analysis tool. Wang et al. [34] introduced DIRE (Diffusion Reconstruction Error), which uses the reconstruction error of a pretrained diffusion model to distinguish generated from real images: generated images can be more faithfully reconstructed by a diffusion model than real ones, producing a discriminative residual.

Our work adapts the underlying principle of DIRE that the reconstruction residual of a diffusion-model denoising operation carries information about the input, but redirects it from the real-vs-generated detection task to the device-attribution task. Where DIRE asks “was this image produced by a diffusion model?”, we ask “does the residual produced by *this camera’s* diffusion model match the residual of *this query patch*?”. The shift requires a different training protocol (per-camera DDPMs trained on real images from each device), a different residual definition (partial-noise / partial-denoise rather than full inversion), and different downstream matcher (NCC against averaged fingerprints, refined via Linear Discriminant Analysis (LDA), rather than binary classification). To our knowledge, this is the first application of diffusion models to source camera identification.

3. Approach

3.1. Pipeline Overview

Our pipeline, as shown in Figure 1, mirrors the five-stage structure of classical PRNU-SCI but replaces the hand-crafted denoising filter with a per-camera trained Denoising Diffusion Probabilistic Model. For a target camera set $\{C_1, \dots, C_N\}$ the procedure consists of (i) training, in which a separate DDPM ϵ_{θ_C} is trained on patches from each camera; (ii) residual extraction, in which a partial-noise / partial-denoise round trip through the trained model produces a device-discriminative residual $R(I) = I - \hat{x}_0$ for each patch I ; (iii) fingerprint construction, in which the mean residual over K reference patches yields a per-camera fingerprint

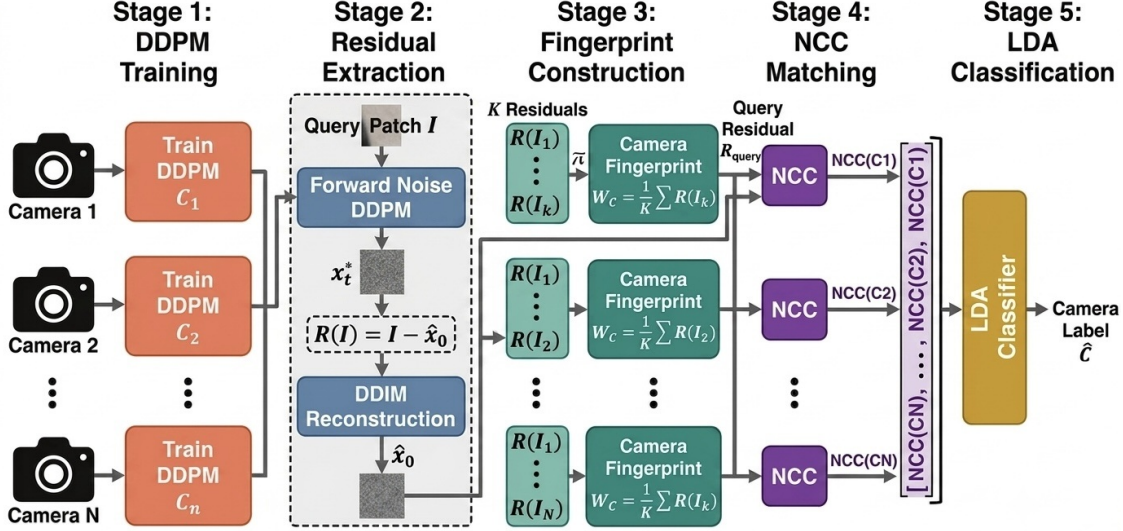


Figure 1. Proposed per-camera DDPM-SCI pipeline. A separate DDPM is trained for each camera; query patches are partially noised to timestep t^* and reconstructed through DDIM sampling under each candidate camera’s model, producing residuals matched to per-camera fingerprints via NCC.

$W_C = \frac{1}{K} \sum_k R(I_k)$; (iv) identification, in which the query residual is matched against each fingerprint via Normalised Cross-Correlation, and (v) LDA refinement, in which the matched residuals are projected via Linear Discriminant Analysis to maximise between-class separation before final camera assignment.

3.2. Diffusion Model Background

A Denoising Diffusion Probabilistic Model [31, 33] defines a forward Markovian process that gradually corrupts a clean sample x_0 with Gaussian noise over T steps:

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I}), \quad (3.1)$$

with a fixed noise schedule $\{\beta_1, \dots, \beta_T\}$. The closed-form expression $x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$, where $\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s)$ and $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, allows direct sampling at any timestep. The reverse process is parameterised by a neural network $\epsilon_\theta(x_t, t)$ trained to predict the injected noise via the standard objective

$$\mathcal{L}_{\text{DDPM}}(\theta) = \mathbb{E}_{x_0, t, \epsilon} \left[\|\epsilon - \epsilon_\theta(x_t, t)\|^2 \right]. \quad (3.2)$$

At inference, the network’s reconstruction of x_0 from x_t is given by $\hat{x}_0 = (x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(x_t, t)) / \sqrt{\bar{\alpha}_t}$. For efficient inference we use DDIM sampling [32], which recasts the reverse process as a deterministic ODE and reduces sampling from $T = 1000$ to $n = 50$ steps at negligible quality cost.

3.3. Network Architecture

The denoising network ϵ_θ , as shown in Figure 2, is a time-conditioned U-Net [37] with three downsampling operations across four resolution levels, operating on RGB patches of shape $3 \times H \times W$ with $H = W = 128$. The encoder applies four residual blocks at successively halved resolutions with channel counts $(C, 2C, 4C, 8C)$ where $C = 64$, downsampling $128 \rightarrow 64 \rightarrow 32 \rightarrow 16$; the bottleneck applies a single residual block at the 16×16 deepest level; and the decoder mirrors the encoder with skip connections concatenating corresponding encoder activations. Spatial downsampling between levels uses average pooling with stride 2, and upsampling uses nearest-neighbour interpolation. Each residual block applies GroupNorm (8 groups), SiLU activation, 3×3 convolution, addition of the timestep embedding, and a residual skip; a 1×1 convolution reshapes the skip when the input and output channel counts differ. The scalar timestep t is encoded via sinusoidal positional embedding, passed through a two-layer MLP, and projected linearly into each residual block as an additive bias. The total parameter count is approximately 20M.

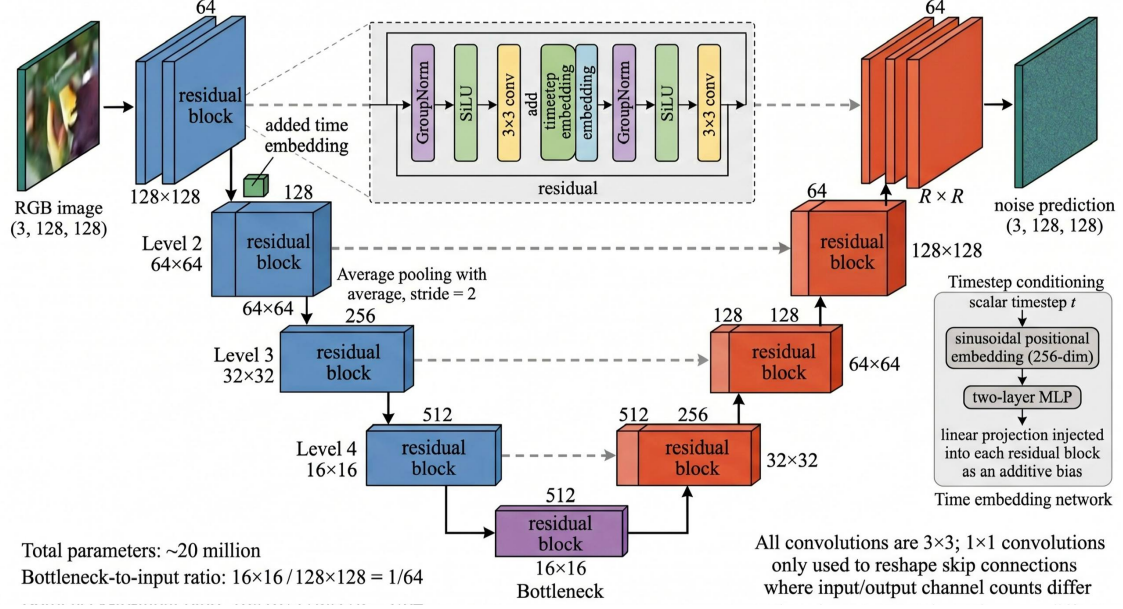


Figure 2. Time-conditioned U-Net used as the per-camera denoiser. Three downsampling operations across four resolution levels (spatial resolutions $128 \times 128 \rightarrow 64 \times 64 \rightarrow 32 \times 32 \rightarrow 16 \times 16$, channel counts 64, 128, 256, 512), with sinusoidal timestep embedding injected via a two-layer MLP into every residual block.

3.4. Residual Extraction and Fingerprinting

At inference, given a clean patch I we add partial noise to reach a chosen timestep t^* :

$$x_{t^*} = \sqrt{\bar{\alpha}_{t^*}} I + \sqrt{1 - \bar{\alpha}_{t^*}} \epsilon, \quad \epsilon \sim \mathcal{N}(0, \mathbf{I}). \quad (3.3)$$

DDIM sampling [32] then deterministically denoises from t^* back to 0 in $n = 50$ steps. After 50 iterations the model produces the reconstruction $\hat{x}_0$, from which the residual is $R(I) = I - \hat{x}_0$.

The choice of t^* reflects a trade-off. Low values produce minimal perturbation, leaving the reconstruction nearly identical to the input and the residual dominated by reconstruction error rather than device signature. High values produce nearly-pure noise, forcing the network to hallucinate content and yielding a residual that reflects scene content rather than device characteristics.

For each camera C , the reference fingerprint is the mean of K residuals computed from training patches of that camera: $W_C = \frac{1}{K} \sum_{k=1}^K R(I_k)$.

3.5. Identification and Matching

For each query patch I_q , residuals are extracted through each of the N camera-specific DDPMs, producing N candidate residuals $R_q^{(C_1)}, \dots, R_q^{(C_N)}$. The NCC score between the query residual and each camera fingerprint forms a feature vector $\mathbf{s}_q = [\text{NCC}(R_q^{(C_1)}, W_{C_1}), \dots, \text{NCC}(R_q^{(C_N)}, W_{C_N})]$, which is passed to the LDA classifier for final camera assignment. Without LDA, the predicted camera is:

$$\hat{C} = \arg \max_{C_i} \text{NCC}(R_q^{(C_i)}, W_{C_i}). \quad (3.4)$$

Both variants (with and without LDA) are reported in all experiments.

4. Experiments

4.1. Datasets and Setup

We evaluate on three benchmark datasets: Northumbria Temporal Image Forensics (NTIF) [38], Dresden [39], and VISION [40], all datasets consisting of ten camera devices and for all three datasets, patches of size

128×128 are extracted with up to 20 patches per image and per-camera caps applied to balance the dataset for training stage. Splits are performed at the image level to avoid leakage, with 70% of images assigned to training, 15% to validation, and 15% to test.

All experiments use the linear noise schedule $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$ with $T = 1000$ diffusion timesteps, extraction timestep $t^* = 500$, and $n = 50$ DDIM sampling steps [32]. Training uses AdamW with learning rate 2×10^{-4} , batch size 128, EMA decay 0.999, and gradient clipping at ℓ_2 -norm 1.0. Each camera’s DDPM is trained independently for up to 100 epochs with early stopping. Hardware is a single NVIDIA RTX 6000 Ada Generation GPU (48 GB VRAM). Code and models are available from the corresponding author upon request.

4.2. Classical Baseline Methods

We compare against the classical denoising Wiener filter used as drop-in replacements for the DDPM in the same pipeline: residual extraction, fingerprint construction via averaged residuals, and NCC matching. The Wiener filter follows the Lukáš et al. [4, 5] formulation with a 3×3 spatially adaptive kernel. We used the same train/test split and fingerprint construction protocol for Wiener filter as the proposed DDPM pipeline.

4.3. Results

Table 1 reports closed-set classification performance across all three datasets at 128×128 patches. The results are presented as a component-wise ablation: the Wiener filter establishes the classical baseline, the DDPM baseline isolates the contribution of the learned residual extractor, and DDPM+LDA adds the discriminant post-processing stage. TPR and TNR are computed in a one-vs-rest scheme across all ten cameras and macro-averaged; Overall = (TPR + TNR)/2 is the macro-averaged balanced accuracy, chosen to account for the asymmetry between the single positive class and nine negative classes in each binary decision.

DDPM baseline vs. Wiener filter. As it can be seen from the Table 1 the DDPM baseline outperforms the Wiener filter on all three datasets. On Dresden, the DDPM baseline raises TPR from 65.65% to 80.67% (+15.02 points) and at the same time improving TNR from 96.64% to 97.54%, yielding an overall gain of 7.97 points (81.14% $\rightarrow$ 89.11%). On VISION, the gain is more modest: TPR improves by 3.80 points and overall accuracy rises from 87.70% to 89.93%. Finally, on Northumbria, TPR improves from 73.53% to 78.79%, yielding an overall gain of 2.95 points (85.28% $\rightarrow$ 88.23%). The consistent TPR improvement across all three datasets demonstrates that a per-camera learned content prior produces residuals that are more device-discriminative than the classical hand-crafted Wiener filter, particularly in terms of correctly attributing query patches to their source camera (TPR). The high TNR values across all methods reflect that the 10-camera closed-set task is relatively easy to reject incorrect cameras once a discriminative residual is available.

Effect of LDA refinement. Applying LDA as a post-processing step consistently improves accuracy across all three datasets and both TPR and TNR metrics. The largest gain is on Northumbria, where LDA raises overall accuracy from 88.23% to 93.74% (+5.51 points), the largest absolute improvement across all dataset–method pairs. On Dresden, LDA raises TPR from 80.67% to 88.99% (+8.32 points) and overall accuracy from 89.11% to 93.84% (+4.73 points). On VISION, the gain is 2.57 points overall (89.93% $\rightarrow$ 92.50%).

The mechanism behind LDA’s benefit is structural. Without LDA, camera assignment is based on the scalar NCC score between the query residual and a single fingerprint. This is a one-dimensional comparison that discards the relational geometry between all N camera fingerprints. With LDA, the full NCC feature vector $\mathbf{s}_q = [\text{NCC}(R_q^{(C_1)}, W_{C_1}), \dots, \text{NCC}(R_q^{(C_N)}, W_{C_N})]$ is projected into a discriminant subspace that maximises between-camera separation relative to within-camera scatter. This allows the classifier to exploit cross-camera correlations — for instance, if two cameras produce systematically correlated NCC scores, LDA can learn to down-weight that dimension. Crucially, LDA is trained on the training split residuals and evaluated on the test split, so its gains are not an artefact of overfitting.

Table 1. Classification performance across three benchmark datasets at 128×128 patches, one centre patch per image for evaluation; up to 20 patches per image used for training. Overall = $(\text{TPR} + \text{TNR}) / 2$.

Dataset	Method	TPR (%)	TNR (%)	Overall (%)
Dresden	Wiener filter	65.65	96.64	81.14
	DDPM baseline	80.67	97.54	89.11
	DDPM + LDA	88.99	98.70	93.84
VISION	Wiener filter	77.87	97.54	87.70
	DDPM baseline	81.67	98.20	89.93
	DDPM + LDA	86.33	98.67	92.50
Northumbria	Wiener filter	73.53	97.04	85.28
	DDPM baseline	78.79	97.68	88.23
	DDPM + LDA	88.72	98.77	93.74

Table 2. Inference, training time, fingerprint estimation, and time per image on the Dresden 10-camera benchmark. Training is the one-time offline cost (per-camera DDPM models). Fingerprint estimation builds one reference fingerprint per camera. Time per image is the per-query inference cost using camera-adaptive deployment.

Method	Training Time (min)	Fingerprint Estimation (s)	Time per image (ms)
Wiener filter	–	≈ 30	280
DDPM baseline	≈ 250	≈ 120	2093
DDPM + LDA	≈ 250	≈ 120	2154

Cross-dataset comparison. Dresden achieves the highest overall accuracy (93.84% with LDA), followed by Northumbria (93.74%) and VISION (92.50%). The ranking likely reflects differences in inter-camera diversity: Dresden consists of eight manufacturers with varied sensor sizes, providing well-separated device signatures; VISION has smartphones with more homogeneous sensor characteristics; and Northumbria, as a temporal forensics dataset, includes repeated captures under varying conditions that may reduce within-camera fingerprint consistency. Across all three datasets, DDPM+LDA outperforms the Wiener filter by a margin of 12.70 points (Dresden), 8.46 points (Northumbria), and 4.80 points (VISION), confirming that the proposed pipeline generalises beyond a single benchmark.

Computational cost. Table 2 reports training, fingerprint estimation, and per-image inference time, illustrating the accuracy-versus-complexity trade-off of the proposed approach. Training the 10 per-camera DDPMs takes approximately 250 minutes in total; fingerprint estimation requires ≈ 120 s per camera. The Wiener filter requires no training and constructs fingerprints in approximately 30 s. Per-image inference for the DDPM baseline is $\approx 2,000$ ms, reflecting the 50-step DDIM sampling pass through 10 camera-specific models; the marginal cost of LDA classification adds ≈ 100 ms ($\approx 2,100$ ms total). The Wiener filter processes each image in ≈ 280 ms. The proposed pipeline therefore trades increased computational cost for substantially improved identification accuracy, a trade-off that is acceptable in forensic investigation contexts, where accuracy is paramount and processing time is not a primary constraint. Further acceleration via reduced DDIM steps or model distillation remains a direction for deployment-constrained scenarios.

5. Conclusions

We have introduced a per-camera Denoising Diffusion Probabilistic Model framework for Source Camera Identification, training a separate DDPM for each camera and a partial-noise / partial-denoise residual replaces the classical Wiener filter. Applied to 128×128 patches across three benchmark datasets, the proposed method achieves macro-averaged balanced accuracies of 93.84%, 92.50%, and 93.74% on Dresden, VISION, and Northumbria respectively, outperforming the classical Wiener filter on all three benchmarks.

Two findings worth highlighting, first, even at the modest 128×128 operating scale, the DDPM residual consistently outperforms the Wiener residual at the patch level, showing that a learned per-camera content prior extracts more device-discriminative signal than a hand-crafted local-mean smoother. This supports the broader hypothesis, motivated by DIRE [34], that diffusion model reconstruction residuals carry information about the specific distribution on which the model was trained, which here is the per-camera image distribution rather than the real-vs-generated distinction.

Second, LDA refinement of the NCC feature vector provides consistent and substantial gains across all three datasets, with improvements of up to 5.51 points (Northumbria). By projecting the full N -dimensional NCC score vector into a discriminant subspace, LDA exploits inter-camera relational geometry that scalar argmax matching discards. This suggests that the NCC feature vector contains structured between-camera information beyond the top-scoring dimension, and that lightweight discriminative post-processing is a simple but effective addition to the residual-based pipeline.

Limitations of the present work include the single patch size evaluated (128×128), the restriction to closed-set identification, the computational overhead of per-image DDIM sampling relative to the Wiener filter, the per-camera training requirement (≈ 25 minutes per camera) which may not generalise directly to large camera populations as each new device requires training a fresh generative model, and results being reported from a single experimental run with variance across multiple runs left for future work. The Wiener filter serves as the primary baseline as it represents the classical PRNU-SCI pipeline that the proposed DDPM directly replaces; the extended journal version of this work will compare with CNN-based SCI techniques and learnt baselines like NoisePrint. Future research will focus on inference acceleration via DDIM step reduction, few-shot or transfer-learning strategies to reduce training overhead, extension to video data, and open-set rejection via NCC.

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